

Title

A Crossover Study of Culinary-Range Turmeric and Pepper Extract Effects on Pain Perception in Midlife and Older Adults with Persistent Pain

Abstract

Chronic discomfort remains a significant public health challenge, with many individuals seeking nutritional adjuncts alongside conventional therapies. This randomised crossover trial will investigate whether commonly-available spice components at levels feasible within typical dietary intake can influence pain perception. Adults aged 40 and above with moderate baseline discomfort will be randomised to receive different formulations in a crossover sequence. Each participant will report daily pain ratings via text-based prompts using a standardised 0-10 scale, with assessments captured three times daily throughout the intervention periods. The study will examine whether certain spice combinations produce detectable changes in reported discomfort relative to placebo or single-ingredient controls. Participants will also log any additional pain relief medications taken.

Analysis will employ mixed-effects regression to account for repeated measurements and individual variation. This design aims to determine whether accessible, food-level interventions warrant further development for pain management.